

Spring-Loaded GaN Packaging for Cryogenic Thermal Cycling

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Abstract— Cryogenic power module packaging brings additional challenges for lack of low-temperature soft potting material and flexible electrical connections. Thermal expansion mismatches among different rigid layers of the package can cause the die failure. This paper develops and demonstrates a cryogenically cooled GaN packaging for a 750V/100A solid-state circuit breaker (SSCB) module. The proposed printed circuit boards (PCB)/direct bonded copper (DBC) hybrid module, clamped by spacers and springs to form a spring-loaded structure, can survive thermal cycling down to cryogenic temperatures. Experimental results of thermal cycling and interrupting a 10X fault current (1kA) by the proposed module are provided to show the effectiveness of the proposed packaging method operated with cryogenic cooling.

Keywords— GaN HEMTs, packaging, cryogenic power electronics, CTE mismatch

I. INTRODUCTION

Cryogenic power electronics have drawn much attention recently due to their high power density, high efficiency, and superior performance in various fields including electric aircraft propulsion, deep space applications, and superconducting machines and systems. [1] Power semiconductor devices play a crucial role in cryogenic electronics performance. Si MOSFETs and Si IGBTs offer reduced on-resistance and faster switching speed at low temperatures; however, their decrease in breakdown voltage can be problematic [2-3]. Wide-bandgap devices, such as silicon carbide (SiC) and gallium nitride (GaN), are emerging as replacements for Si devices due to their superior characteristics. Notably, GaN High Electron Mobility Transistor (HEMT) is reported to offer higher conduction loss reduction at cryogenic temperature ($<153^{\circ}\text{C}$) compared to both Si and SiC devices [4]. These advantages are highly beneficial for cryogenic power electronics. Several converters have been reported to operate at cryogenic

temperatures [5-8]. However, these works mainly rely on discrete devices, leading to low power density and higher loop inductance.

A power module can offer a more compact design compared to a discrete device-based design. However, there are no off-the-shelf GaN power modules that work for cryogenically cooling applications. Traditional wire-bond power modules are prone to failure under severe thermal cycling [9]. Additionally, wire bonds are considered the main source of stray inductance [10], which can cause transient overvoltage in GaN devices during current commutations. Moreover, the typical encapsulant used in power modules, such as silicone gel, would become hard and brittle therefore unreliable at cryogenic temperature [11].

Printed circuit board (PCB) embedded designs can minimize the parasitic inductance through the layout optimization. However, the dielectric layer in PCBs has poor thermal conductivity. Using a thinner dielectric layer can mitigate this issue, but it may result in high electric field stress and potential breakdown. A PCB/ direct bonded copper (DBC) hybrid module has emerged as a solution to optimize both parasitic inductance and thermal resistance [12-15]. When adopting this approach for the cryogenic application, GaN dies fail to survive thermal cycling due to thermal-mechanical stress related to the properties of packaging materials. An improved packaging method is then highly demanded.

Thus, this work presents a cryogenically cooled enabled power module with GaN HEMTs bare dies and improved PCB/DBC hybrid packaging with a proposed spring-loaded structure. The packaging method is implemented and verified for a 750V/100A solid-state circuit breaker (SSCB) module, although the same technique can be easily applied to power module designs for various cryogenic power electronics applications.

II. FAILURE OF PCB/DBC HYBRID GAN PACKAGING IN CRYOGENIC APPLICATIONS

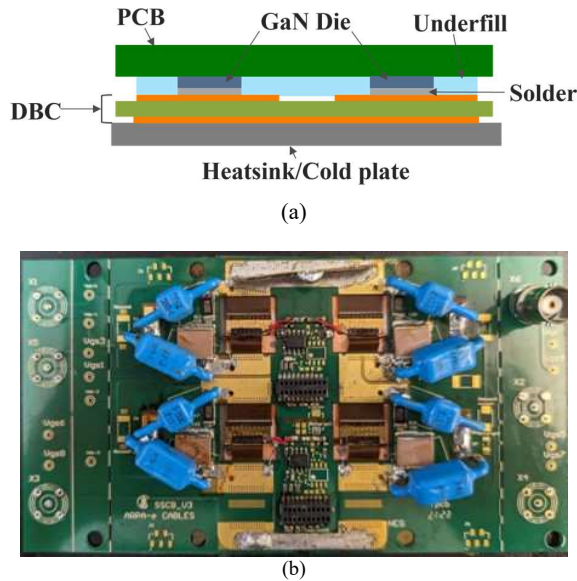


Fig. 1. Exemplary structure of a PCB/DBC hybrid packaging (a) structure of PCB/DBC hybrid packaging (b) sample SSCB module

In cryogenic power electronics applications, the power module must withstand more severe thermal cycling, e.g., from room temperature (23°C) down to cryogenic temperature (-153°C) or even lower to liquid nitrogen temperature (-195°C), yielding at least 170°C difference. The wide temperature operating range and different material coefficients of thermal expansion (CTE) can result in extra stress at joints of two or more different materials. In addition, the material would have different characteristics at extremely low temperatures, which could affect the reliability of the power module.

Typical PCB/DBC hybrid packaging from top to bottom consists of PCB, devices, and DBC. Dies are soldered/sintered and sandwiched between the PCB and DBC. Gate driver and decoupling capacitors are mounted on the PCB to reduce the loop parasitics and enhance electrical performance such as increased switching speed, reduced voltage/current ringing, and alleviated voltage stress on devices. On the bottom side, a DBC provides high thermal conductivity and electrical isolation. However, the dies in such structures can potentially crack when the temperature swings from the room temperature to cryogenic temperature and vice versa, caused by the CTE mismatch among materials. The exemplary structure of PCB/DBC hybrid

TABLE I
THE COEFFICIENT OF THERMAL EXPANSION OF SELECTED MATERIALS

Materials	CTE (ppm/°C)
PCB (FR4)	17-19
Copper	16
DBC (AlN and copper)	4.7
GaN die	3.2-5.6
Solder Paste (63Sn-37Pb)	23.9
Aluminum	21 - 24

packaging is built and shown in Fig. 1. PCBs are four-layer boards with conventional FR4 materials. Four 650 V/150 A GaN HEMTs, GS-065-150 bare dies are used. Active regions of dies are soldered to the PCB with substrates soldered to the DBC. Full modules are then placed on an aluminum cold plate with cryogenic heat-transfer greases in between. The CTE of selected materials for the PCB/DBC hybrid power module is listed in TABLE I. The large temperature drops from room to

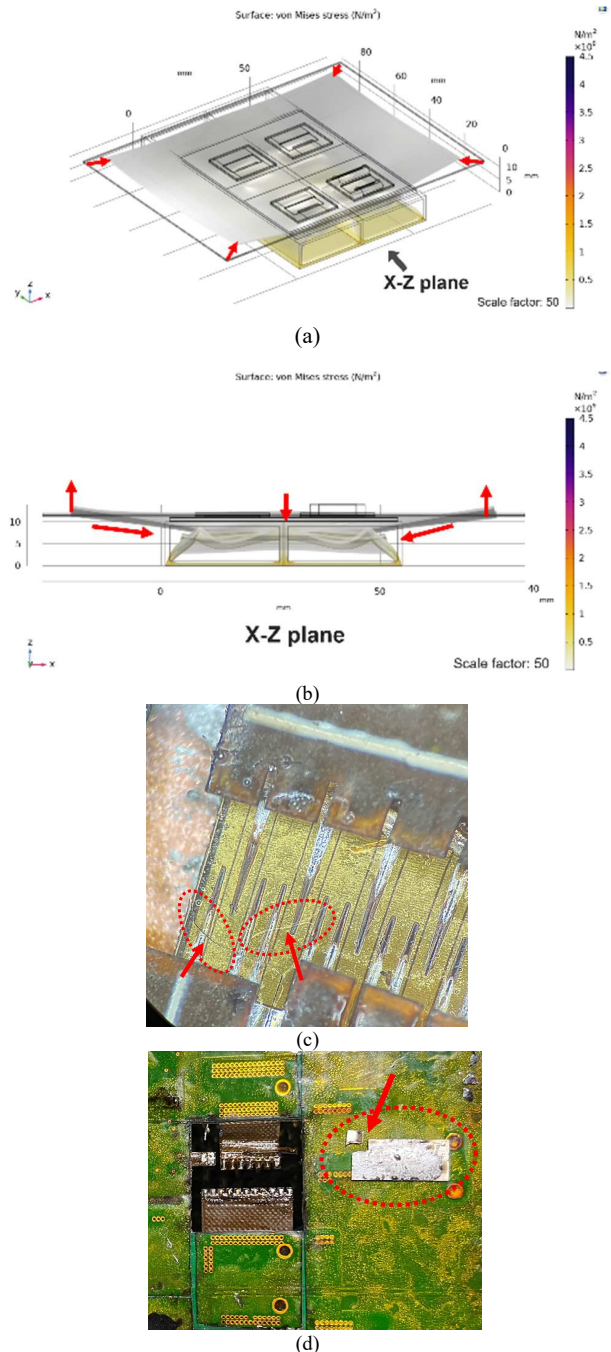


Fig. 2. Deformation trend and cracks observed after thermal cycling (a)(b) deformation trend in simulation, (c) GaN die crack observed under microscope and (d) normal view.

cryogenic temperature and the different CTEs of PCB, GaN bare dies, solder, and DBC result in different contractions or expansions in the process of thermal cycling. The deformation would happen in the process of temperature drops as simulation results shown in Fig. 2(a). The condition of the simulation is set to have temperature drops from the room temperature to -180°C. The contraction of materials can be observed more clearly with an intentionally set 50x scale-up. Since the PCB and copper have higher CTEs than GaN dies and DBC, the large contraction results in an upward bending shown in Fig. 2(b). The mechanical strain generated by the deformation at the interfaces of dies-PCB and die-DBC results in damage to the bare dies, as shown in Fig. 2(c)(d).

III. PROPOSED SPRING-LOADED APPROACH

To prevent the mechanical strain on the interface, the proposed wire-bond-less approach for the cryogenically cooled SSCB module is shown in Fig. 3. Here, the bare dies are not soldered on the DBC. Instead, soft materials such as thermal pads are inserted to relieve the stress of interconnections. This will essentially avoid rigid coupling between PCB and DBC through dies. However, due to insufficient soldering, there could be air gaps among the layers, leading to a poor thermal interface to the cold plate. To ensure good thermal contact without soldering dies on the DBC, all the layers need to be pressed together with a plastic spacer and screws.

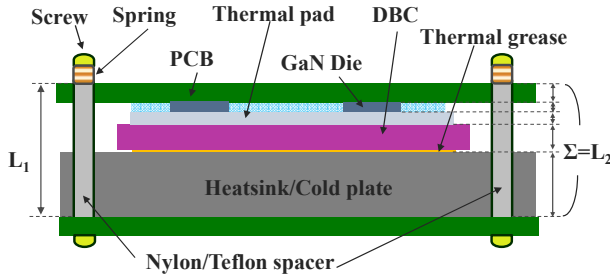


Fig. 3. Structure of proposed spring-loaded spacer PCB/DBC hybrid packaging

Assume L_1 to be the length of the nylon spacer, and L_2 is the overall thickness of dies, thermal pad, DBC, and cold plate. The changes in lengths are given by

$$\Delta L_1 = CTE_{Nylon} \cdot L_1 \cdot \Delta T \quad (1)$$

$$\Delta L_2 = (CTE_{Die} \cdot L_{Die} + CTE_{pad} \cdot L_{pad} + CTE_{DBC} \cdot L_{DBC} + CTE_{Coldplate} \cdot L_{Coldplate}) \cdot \Delta T \quad (2)$$

Where $CTE_{Nylon} = 50-90 \text{ ppm/}^\circ\text{C}$, $CTE_{Die} = 3.2-5.6 \text{ ppm/}^\circ\text{C}$, $CTE_{pad} = 11-180 \text{ ppm/}^\circ\text{C}$, $CTE_{DBC} = 4.7 \text{ ppm/}^\circ\text{C}$, $CTE_{Coldplate} = 24 \text{ ppm/}^\circ\text{C}$ (aluminum), and $\Delta T = 200 \text{ }^\circ\text{C}$.

To ensure no gaps occur during contractions caused by temperature drops, ΔL_1 should be greater than ΔL_2 . However, to reduce the stress caused by the spacer to the module, the difference $\Delta L_1 - \Delta L_2$ should also be minimized. With the nylon

spacer, ΔL_1 could be much larger than ΔL_2 due to the relatively large CTE of nylon. For given conditions, the mismatch between ΔL_1 and ΔL_2 could be 100-200 μm . To alleviate this stress, spring-loaded nylon spacers are introduced to clamp the whole module on the heatsink, or cold plate in this case. The spring is half-compressed at room temperature. In case a gap between layers during thermal cycling, the spring will expand and remove the gap. On the other hand, if the expansion of layers is more than spacers, the spring will compress to avoid additional mechanical stress. Thus, by introducing the spring on the spacer, the mechanical stress or strain caused by the CTE mismatch among materials can be relieved with zero gap between layers being ensured.

IV. EXPERIMENTAL RESULTS

The proposed structure is implemented in an SSCB as shown in Fig. 4. It consists of two dies in parallel and then two in series. Each device is paralleled to a snubber and TVS diode for the current interruption function. Gate drivers are also embedded to achieve compact design and high power density. The module can also be transformed into a half-bridge module by simply removing the snubber circuits and TVS diode.

The SSCB module is designed to undertake 750V/100A, and capable of interrupting a fault current up to 1 kA. The built module is placed on the cold plate and held by nylon spacers with springs, as shown in Fig. 5. The cryogenic thermal grease

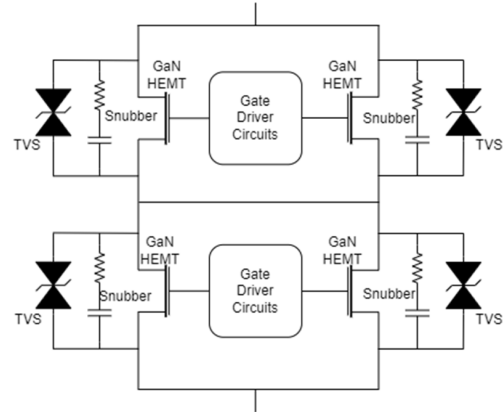


Fig. 4. SSCB module configuration

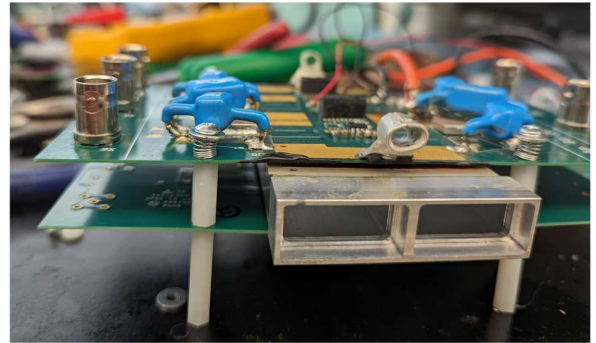


Fig. 5. Proposed spring-loaded packaging SSCB module



Fig. 6. Temperature chamber for thermal cycling

is used between the bottom of the DBC and the cold plate. The module is tested through thermal cycling in a temperature chamber shown in Fig. 6. The thermal cycling starts from room temperature down to -180°C , soaks for 5 minutes before rising back to room temperature and soaks again for another 5 minutes. The thermal cycling rate is about $15^{\circ}\text{C}/\text{min}$ for heating up, and up to $36^{\circ}\text{C}/\text{min}$ for cooling down based on temperature chamber capability. Overall, 5 thermal cycles have been done. The die condition is examined with a curve tracer Keysight B1505A by checking $I_{\text{DS}}\text{-}V_{\text{DS}}$ characteristics and the leakage current, as shown in Fig. 7. The $I_{\text{DS}}\text{-}V_{\text{DS}}$ characteristics remain the same after thermal cycling indicating no die damage in the process. The leakage current of devices also shows no differences before and after thermal cycling, suggesting that no structural damage on the GaN dies and still can withstand standoff voltage.

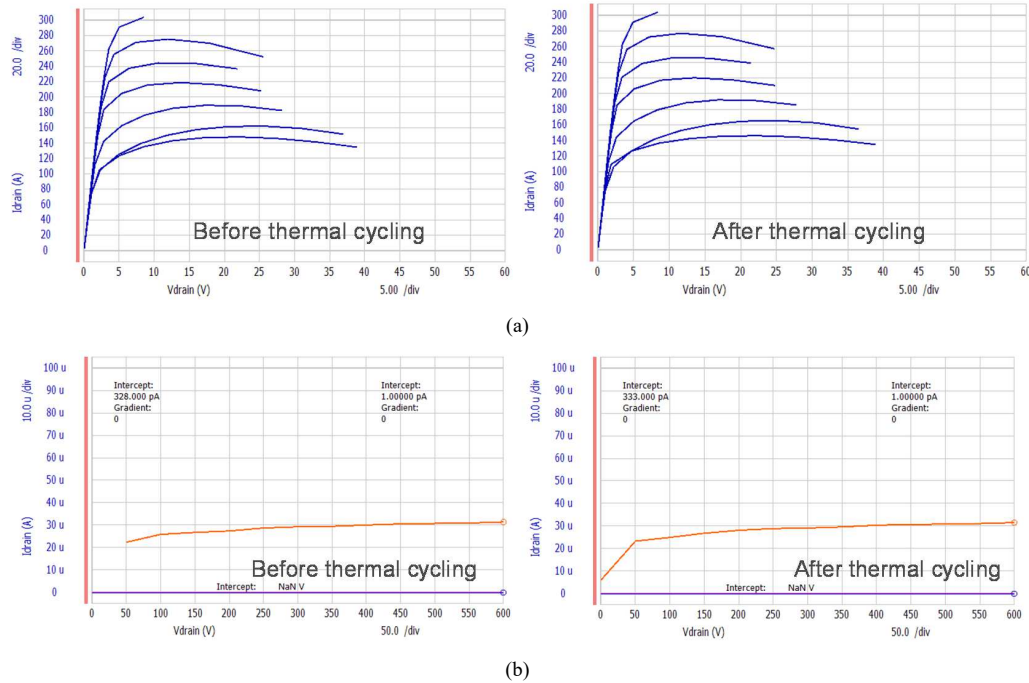


Fig. 7. GaN die characteristic before and after thermal cycling (a) I-V characteristic (b) leakage current

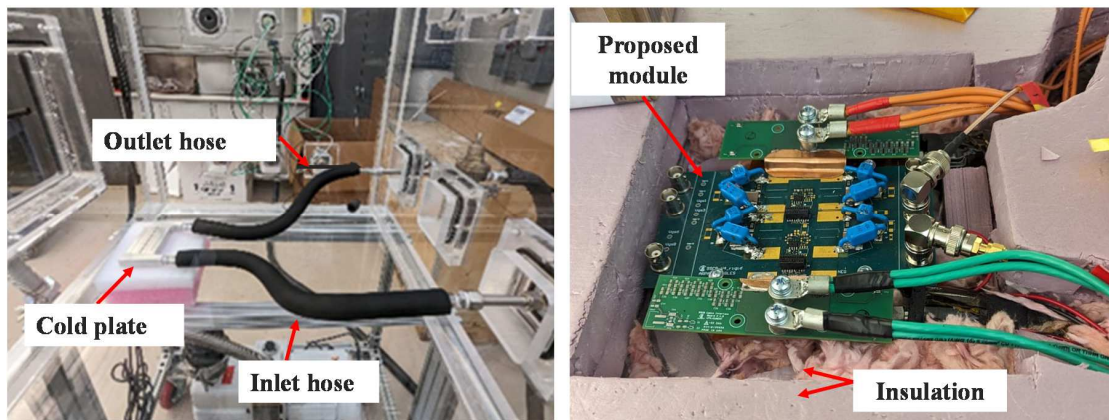


Fig. 8. Cold plate cooling test setup

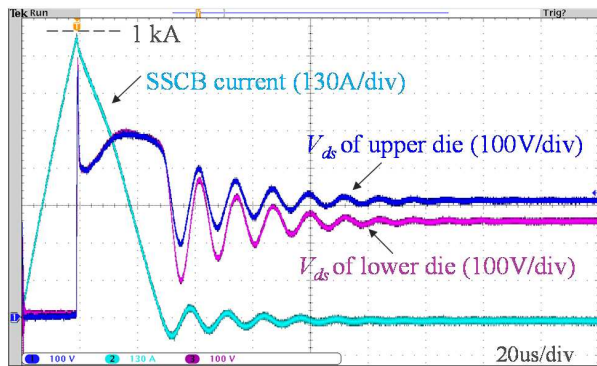


Fig. 9. High fault current interruption results with cryogenic cooling

The SSCB module was also tested on a cold plate with liquid nitrogen cooling, as shown in Fig. 8. The switching-off test is evaluated after liquid nitrogen cools the cold plate down to -190°C . Fig. 9 shows the module capability of interrupting 10x fault current, which is 1kA with liquid nitrogen cooling. The results prove the spring-loaded structure is effective in surviving harsh thermal cycling and the module can still function well.

V. CONCLUSION

An improved packaging method is proposed for the cryogenically cooled application. A spring-loaded structure is introduced to help relieve the thermal-mechanical stress induced by the CTE mismatch of packaging materials. The proposed module survives thermal cycling from room temperature to cryogenic temperature. Experimental results also validate the module's effectiveness of interrupting the high fault current under the cryogenic environment.

Acknowledgment

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